SEMANTICALLY VERIFIED SCIENTIFIC WORKFLOWS

PAPER 2 | Research conception, SciGuard software, experiments, manuscript, and publication strategy

Working manuscript: Beyond Syntactic Validity: Scientific Semantic Constraint Checking for AI-Generated Neuroimaging Workflows

Research protocol and literature-grounded planning document | 25 October 2026 | Version 1.0

Status: proposed research. Neither the method nor the benchmark results described here have been established experimentally. Journal targets are conditional, not acceptance predictions.

Executive decision

Build SciGuard as a useful open-source preflight validator, then evaluate a narrow, falsifiable hypothesis: explicit domain-semantic contracts and deterministic checking can reduce scientifically invalid workflow acceptance compared with file/schema validation and LLM-only review, while maintaining high acceptance of valid workflows. Start with structural connectomics, where coordinate spaces, transforms, atlas labels, interpolation, and provenance are well-defined sources of downstream errors.

Dimension	Proposed decision
Primary paper	Methods paper: scientific semantic verification of computational workflows
Public tool	SciGuard: CLI, Python API, optional service, machine-readable reports
First domain	T1w + diffusion MRI structural connectomics (DK/DKT atlas chains)
Core evaluation	Matched valid/invalid workflow corpus; independent expert reference labels
Main endpoint	Unsafe acceptance rate of invalid workflow specifications
Stretch journal	Nature Methods if cross-domain and biological significance; Nature Computational Science if general formal computational contribution
Strong focused journal	Imaging Neuroscience; IEEE Transactions on Medical Imaging if substantial novel imaging method

Reader's map

- I. Scientific rationale and research questions
- II. Landscape and critical literature review
- III. SciGuard: public tool specification and architecture
- IV. Semantic model and technical methodology
- V. Data, benchmark design, experiments and statistics
- VI. Study execution, governance, risks and reproducibility
- VII. Manuscript package and journal targeting
- VIII. Resource plan, roadmap, release gates and references

I. SCIENTIFIC RATIONALE AND CONTRIBUTION

1. The problem: execution is not scientific validity

Scientific pipelines connect tools whose outputs may be syntactically acceptable yet semantically inappropriate. A NIfTI file may contain a T1-weighted image, an atlas parcellation, a diffusion scalar, or a lesion mask; identical file extensions do not imply interchangeable scientific meaning. A structural connectome pipeline can complete successfully even when atlas labels and tractography occupy incompatible coordinate spaces. Depending on the software and data, the result may be a plausible-looking but invalid connectivity matrix.

Existing workflow engines, BIDS schemas, descriptors, and containers significantly improve interoperable computation. The proposed contribution is NOT the claim that these systems lack type checks or validation. It is the empirical question of whether explicitly modeled domain semantics and provenance-aware constraints add measurable coverage for a defined set of scientific errors beyond selected existing validators and tool descriptions.

2. Primary research question

RQ1: For neuroimaging workflows, how accurately can semantic contracts and deterministic checking classify scientifically valid, invalid, and indeterminate compositions relative to conventional validation and LLM-only review?

3. Secondary research questions

- RQ2: Which constraint families contribute most to safe rejection of invalid workflows?
- RQ3: Does integrating the checker into agent planning reduce invalid accepted/executed workflows?
- RQ4: Can actionable repair suggestions improve correctness without silently introducing new errors?
- RQ5: How does performance change on held-out tools, atlases, site-specific metadata, and task structures?
- RQ6: What are annotation burden, runtime overhead, coverage limitations, and false rejection costs?

4. Proposed hypotheses (to preregister, not assumed true)

- H1: Enriched semantic checking reduces false acceptance of invalid workflow compositions compared with a file/schema baseline.
- H2: A hybrid agent + semantic checker reduces invalid final workflow submissions compared with agent-only review.
- H3: Explicit transform-direction and atlas-identity rules account for a substantial share of the improvement in structural connectomics.
- H4: Benefits persist under tool/task holdouts with only contract-level onboarding, not manually tuned prompts.
- H5: Validation overhead is small relative to the avoided cost of wasted or scientifically invalid pipeline runs.

5. Proposed contributions, conditional on experiments

1. A domain-semantic representation for scientific data artifacts, transformations, tool preconditions, and scientifically meaningful dependencies.
2. A deterministic validator with explainable outcomes: PASS for assessed rules, FAIL, UNKNOWN, and NEEDS_REVIEW; never a blanket scientific-correctness certificate.
3. A controlled benchmark with expert-reviewed positive and negative examples, including realistic and synthetic failure cases.
4. A reproducible comparison to existing interfaces, validators, LLM review, and hybrid systems; ablations by rule family.
5. An independently usable SciGuard release integrated with an existing workflow engine or NeuroInsight, with open examples and documented boundaries.

6. Scope boundaries

Initial scope is compositional validity of specified MRI workflows, not clinical diagnostic validity, image-quality certification, reproducibility of all downstream statistics, or proof of correctness of arbitrary scientific inference. Clinical use and patient-management decisions are outside this first release. A larger life-science methods claim requires additional validation domains.

II. PRIOR WORK AND LITERATURE REVIEW

7. Literature review methodology and limitations

This document is a curated, source-linked scoping review rather than an exhaustive systematic review. The final manuscript should reproduce searches in PubMed, Google Scholar/OpenAlex, IEEE Xplore, ACM DL, arXiv and major standards documentation, recording date, query, inclusion criteria, and screening results. Priority search groups: scientific workflows + semantic validation; neuroimaging coordinates + BIDS derivatives; AI agents + workflow generation + verification; provenance + executable research objects; static type checking + SMT verification.

8. Closest existing systems: critical comparison

Existing work	Established capability / why it matters	What remains to test
Nipype (2011) [1]	Typed interfaces; executable DAGs; connections, provenance, execution abstractions.	Whether extra domain-semantic contracts detect domain errors outside existing interface validation.
BIDS / BIDS Apps [2,3]	Conventions, dataset metadata, app interoperability, dataset and derivative validity.	Cross-tool semantic compatibility, transform direction, and meaning beyond BIDS conformance.
Boutiques (2018) [4]	Rich JSON command descriptors, dependencies, invocation validation, portability.	Added benefit of scientific meaning over invocation/input-constraint checks.
CWL [5]	Portable tool/workflow schemas, type-checkable tool connections.	Ontological and provenance-dependent constraint checking not expressible in basic data types.
Nextflow/Snakemake [6,7]	Production-grade scheduling, caching, dependency management, reproducibility.	Complement, not replace; gate scientific composition before execution.
MCP/OpenAPI [8,9]	Tool exposure and machine-readable schemas for agents/services.	Whether semantic contracts significantly improve agent decision-making over generic schemas.
RO-Crate / W3C PROV [10,11]	Machine-readable provenance and workflow execution records.	Use provenance as evidence for transforms and processing status.
Formal methods/Z3 [12]	Constraint solving and consistency checks.	Useful implementation option; not itself novel.
Paper2Agent / agentic workflows [13,14]	Research-code-to-agent and natural-language-to-workflow orchestration.	Empirical scientific-semantic assurance and explicit UNKNOWN states.

9. What would NOT be novel

- Adding an LLM chatbot to a workflow engine.
- Writing YAML or JSON descriptions for existing command-line tools.
- Validating BIDS directory structure or generic JSON schemas.
- Using Z3 to check logical constraints without an original scientific representation or evaluation.
- Running FreeSurfer, QSIPrep, or MRtrix3 through an agent.

- Showing a handful of examples where obvious incorrect workflows are rejected.

10. Candidate gap and defensible positioning

Scientific artifacts can require dependent semantic types: modality, measurement units, atlas identity, native/reference coordinate spaces, transform ancestry, interpolation regime, acquisition protocol, and evidence-backed QC states. These are not adequately captured by extension-only compatibility or unconstrained language-model judgment. This is a candidate gap, not a proven novelty claim: demonstrate it via carefully specified comparators and a structured systematic review. A convincing contribution must include previously unsupported failure classes, low false-positive rates, and out-of-distribution reliability.

11. Essential domain literature to add during full review

- BIDS and BIDS Derivatives specification: spatial references, derivative naming, provenance fields, and transform representation.
- MRtrix3 connectome construction and label-to-streamline mapping documentation; identify precisely which image-to-track transform conventions are expected.
- FreeSurfer atlas/annotation definitions and voxel-to-world conventions; test DK vs DKT versus aseg confusion.
- ANTs/FSL registration and interpolation guidance, including label-preserving interpolation and transform direction.
- MRtrix3/ACT, QSIRecon and QSIPrep documentation for valid preprocessing dependencies.
- Research on scientific workflow ontologies and ontology-aware composition; include semantic-web and bioinformatics workflow validation literature.
- Design-by-contract, refinement/dependent types, abstract interpretation, and static-analysis frameworks for a principled constraint model.

III. SCIGUARD: PUBLIC TOOL DESIGN

12. Users, tasks and release surface

User	What SciGuard does	Important boundary
Pipeline developer	Checks declared scientific interfaces and connection edges before execution	Cannot infer missing metadata as valid
Research engineer	Runs preflight on connectomics workflows or CI pull requests	Does not replace visual or quantitative QC
AI agent	Uses machine-readable validation and repair diagnostics	Agent cannot bypass hard rejection policies
Research group	Produces auditable validation summary and provenance requirements	Not a clinical product or regulatory clearance

13. MVP features (release 0.1)

- Read declarative workflow DAGs in a versioned JSON/YAML schema.
- Validate declared artifact scientific types and compatible input requirements.
- Check coordinate spaces, atlas/label identity, modality, allowed interpolation, transform provenance and dependency ordering.
- Return PASS/FAIL/UNKNOWN/NEEDS_REVIEW per rule and machine-readable evidence.
- Generate explainable reports with exact violated predicates and suggested manual next steps.
- Expose CLI (`sciguard validate workflow.yaml --format json`), Python API and regression tests.
- Package curated known-valid, known-invalid and indeterminate fixtures; clearly document coverage.

14. Version 0.2–1.0 features

- Preflight integration with NeuroInsight and optional Nipype/CWL importers.
- Policy profiles by workflow domain, e.g. DWI connectome versus structural morphometry.
- Transform-chain validation and metadata-propagation checks.
- Safer candidate repair proposals with human approval and post-repair revalidation.
- RO-Crate/provenance export, issue reports, CI integrations, hosted docs, optional read-only demo.
- External contribution process and rule provenance linked to expert documentation.

15. Proposed repository organization

```
sciguard/
  schema/          # versioned scientific types and workflow schema
  contracts/       # tool operation descriptions
  rules/           # typed predicates, traceable documentation
  adapters/        # BIDS, Nipype/CWL, NeuroInsight inputs
  validation/      # graph checks and status propagation
  reports/         # JSON and human-readable diagnostics
  fixtures/        # positive, negative, unknown examples
  benchmark/       # evaluation tools (separately versioned)
  docs/            # developer and scientist guides
```

```
tests/              # unit/property/integration/regression tests
```

16. CLI/API contract

```
sciguard validate workflow.yaml --policy dwi-connectome --report report.json

sciguard explain RULE-SPACE-002

sciguard check-contract contracts/atlas-resample.yaml
```

```
from sciguard import validate_workflow

result = validate_workflow("workflow.yaml", policy="dwi-connectome")

assert result.status in {"PASS", "FAIL", "UNKNOWN", "NEEDS_REVIEW"}
```

These interfaces are product specifications, not implemented commands in the current study.

17. Safety design

- Read-only preflight by default; no implicit command execution or file modification.
- Fail closed or require explicit review for safety-critical unknown constraints; distinguish policy from scientific evidence.
- Every result records applicable checks and checks not performed.
- Do not accept raw clinical data into a public hosted demo; use local-first architecture and public/synthetic fixtures.
- No LLM-based repair applied automatically to user data; show diff/plan and require authorization.
- Version rule definitions and policy profiles; record software/container digests and provenance.

IV. PROPOSED SEMANTIC MODEL AND METHODS

18. Formal objects

Represent workflow G=(V,E), with V = tools/operations and E = artifact connections. Each artifact a has a semantic descriptor tau(a) with modality, representation, reference space, coordinate convention, atlas, label ontology, units, acquisition/preprocessing state, transform ancestry and relevant QC evidence. An operation v has input predicates P_v, output predicates Q_v, transformation effect T_v, and explicit limits. A connection (a,v) is accepted only when required predicates are established, rejected when a contradiction is demonstrated, and marked unknown otherwise.

The representation should use three- or four-valued results instead of treating absent information as false or true. Rule evaluation is epistemic: known compatible, known incompatible, insufficient evidence, or policy-required expert review.

19. Constraint categories and examples

Constraint family	Example predicate	Expected validation action
Spatial	atlas.space == streamline.reference_space OR validated_transform_chain	Reject direct mismatch; inspect transform evidence
Atlas/ontology	atlas.label_set compatible with expected node definition	Reject incompatible dictionary/label convention
Interpolation	discrete label image must not use continuous-valued interpolation without documented justification	Flag/require review

Pipeline state	tractography requires compatible FOD/model and preprocessing	Reject missing declared preconditions
Acquisition	method prerequisites consistent with DWI shells/directions and metadata	Reject or unknown depending on provenance
Units	connectome weight units or normalization interpretation declared	Flag incompatible comparisons
Lineage/provenance	registration transform derived from same subject/session/reference pairing	Reject cross-subject evidence
QC	required QC gate exists and documented threshold checked	Unknown/review; not universal validity claim

20. Example scientific contract

```
schema_version: 0.1
operation: map_labels_to_diffusion
inputs:
 labels: {kind: parcellation, atlas: DesikanKilliany,
        space: subject_T1w, value_type: discrete_labels}
 reference: {kind: diffusion_reference, space: subject_DWI}
 transform: {kind: spatial_transform,
           from_space: subject_T1w, to_space: subject_DWI,
           same_subject: true, verified: true}
preconditions:
 - subject_identity_consistent
 - transform_direction_valid
 - label_preserving_interpolation
outputs:
 labels_dwi: {kind: parcellation, atlas: DesikanKilliany,
           space: subject_DWI, value_type: discrete_labels}
validation_policy: reject_on_known_mismatch
```

The structure is hypothetical: it is not an official schema from BIDS, FreeSurfer, MRtrix3 or CWL.

21. Soundness and completeness: precise claims

Do not claim global soundness for scientific validity. At most prove properties within a formalized subset (e.g., if all declared reference-space predicates are correct and mandatory transformation evidence is complete, a checker will reject connections with contradictory spaces). The scientific assumptions underlying annotations, reference frames and operator semantics require independent empirical validation. A sound rule can be undermined by erroneous metadata.

22. Algorithms

6. Normalize tool and artifact descriptors to canonical semantic types; preserve source and version for each asserted field.
7. Perform graph structure and cycle/dependency checks using the workflow engine's existing facilities.
8. Propagate known semantic attributes and transform effects along typed edges.
9. Evaluate explicit constraints, recording missing evidence separately from contradiction.
10. Optionally use an SMT solver for cross-edge constraints; start with deterministic predicates, add solver only if complexity warrants.
11. Emit findings with traceable evidence, rule version, severity, confidence classification (not LLM confidence), and suggested investigation.
12. For proposed repairs, re-run the complete check and require explicit user authorization for any execution.

23. Why the method might add value

Consider an output that is a valid NIfTI integer image and an input requiring a valid NIfTI integer image. Generic file and schema typing can succeed while semantic space and atlas identity do not match. SciGuard is designed to surface the mismatch. Its value must be demonstrated empirically against well-configured baselines that use all metadata and validation already provided by the existing tools.

V. DATASETS, BENCHMARK AND EXPERIMENTS

24. Acquisition plan

Dataset/fixture	Purpose	Access/constraints
Synthetic image and transform fixtures	Exact ground truth for axis, units, spaces, atlas, label interpolation	Generated; openly redistributable
BIDS examples repository	Directory/metadata structure and schema conformance	Examples may omit actual imaging voxel data
OpenNeuro ds000114	Public T1w and diffusion examples; test/retest contexts	Check current snapshot/license/availability
Additional OpenNeuro BIDS dataset	Held-out scanner/protocol/task transfer	Select only after audit of raw modalities and license
Curated expert workflows	Positive reference configurations from public documentation	Permission/copyright and version citation required
Corrupted/perturbed workflow variants	Controlled negative tests, including plausible silent errors	Mutation provenance and expert adjudication

25. Benchmark composition and sampling

A useful planning target is 240 workflow configurations: 80 independently authored valid cases, 120 invalid variants distributed across scientific error classes, and 40 deliberately ambiguous/insufficient-evidence cases. This is a planning value—not a power calculation. Use a 40–60 task pilot to estimate detection rates, disagreement and error-category prevalence, then choose the final size using confidence-interval precision and paired-study assumptions.

Avoid treating mutations of the same parent workflow as statistically independent. Split by source workflow, acquisition family, atlas/tool chain and authorship, not simply by individual file. Reserve a held-out group of tools and workflow structures and a frozen test set prior to selecting rules or prompts.

26. Error mutation library

Mutation	How generated	Gold-standard review
Space flip	Change labels from T1w to DWI without transform	Expert reviews whether connection is truly invalid
Transform direction	Reverse from/to spaces for the declared mapping	Check effect and software transform conventions
Subject/session swap	Attach transform from a different subject or session	Confirm provenance contradiction
Atlas ID swap	Change DK to DKT while retaining incompatible label interpretation	Check exact atlas table and downstream assumptions
Interpolation change	Resample discrete labels with unsuitable mode	Validate against tool and use-case requirements
Dependency deletion	Remove declared processing step/model prerequisite	Ensure required step not hidden elsewhere
Metadata omission	Remove key coordinate or acquisition descriptors	Gold label UNKNOWN rather than assume invalid
Normalization inconsistency	Claim incomparable edge-weight semantics	Check analysis definition and units

27. Ground truth and annotation protocol

13. Write a task specification with inputs, intended scientific goal, expected output, permissible alternatives and evidence sources.

14. Two independent neuroimaging reviewers label validity, error family, severity, and evidence sufficiency; use a third adjudicator for disagreements.
15. Report agreement by class (Cohen kappa or appropriate alternative) and adjudication frequency; retain an uncertainty class.
16. Do not ask reviewers to judge a proposed method only after seeing SciGuard outputs; hide system predictions during adjudication.
17. Publish anonymized fixture IDs, mutation seeds, and rules for generating every case, respecting any third-party data constraints.

28. Comparators

System	Description	Fairness consideration
B0: Existing schema/engine	BIDS Validator + selected workflow engine's ordinary type/input checks	Configure correctly; document precisely checked constraints
B1: Tool-specific validations	Tool native validation (e.g., Nipype interface checks, command invocation validation)	Do not omit native checks to inflate gain
B2: LLM scientific review	Agent examines docs, workflow manifest and available metadata	Same task evidence and time/token budget
B3: Rules-only semantic validator	SciGuard deterministic predicates without LLM	Isolate rule-engine contribution
B4: LLM + SciGuard	Agent planning with explicit checker feedback and policy	Same execution resources and model settings
B5: Strong expert baseline	Expert-curated workflow where feasible	Human time and task scope reported

29. Experiment matrix

E1 - Offline classification

Evaluate valid, invalid and unknown workflow specifications without running costly image processing. Report macro/micro measures by error family.

E2 - Agent composition

Give agents research goals and a capability catalog. Evaluate final submitted plans with and without SciGuard enforcement, blind-scored by reference checks.

E3 - Error diagnosis and repair

Ask systems to identify error causes, propose minimal repairs, and revalidate. Separately rate necessary changes, unsafe repair and overcorrection.

E4 - Out-of-distribution transfer

Hold out atlas definitions, tools, dataset metadata regimes, and workflow structures. Avoid training rules on held-out instances.

E5 - End-to-end materiality

Select a smaller, curated set of cases where workflows can execute and quantify downstream QC/connectome impact from errors and prevention.

E6 - Overhead and usability

Measure time, memory, false-positive investigation load, human interventions, and annotator/developer time.

30. Primary and secondary endpoints

Primary: unsafe acceptance rate = invalid configurations accepted as executable / all reference-invalid configurations (with confidence interval). Also report valid acceptance rate to expose overconservative policies. UNKNOWN/NEEDS_REVIEW should be analyzed explicitly: coverage, abstention rate, and retained error risk among auto-accepted cases.

Secondary: sensitivity and specificity under predeclared treatment of unknowns; per-error-class recall; false rejection; repair success; execution success; scientific output accuracy on selected executed cases; tokens/cost; decision latency; cross-domain generalization; expert review burden.

31. Statistical analysis plan

- Task pairing: use the same held-out configurations across methods and fixed random seeds where possible.
- Binary paired differences: McNemar or paired bootstrap; cluster by parent workflow and dataset/source to avoid pseudo-replication.
- Uncertainty: 95% confidence intervals for primary rates and effect differences; avoid overinterpreting p-values.
- Multiple hypotheses: identify one primary contrast (e.g., B0 vs B3 unsafe acceptance) and adjust planned secondary families.
- Subgroup analyses: error category, missing metadata, model family, task complexity, atlas, tool holdout.
- Models: at least two agent families where feasible; lock prompts, temperature/decoding, versions and budgets.
- Predefine treatment of abstentions. Do not silently score UNKNOWN as success or failure across different experiments.
- Sensitivity analyses: vary severity weighting, unknown policy, task source, and annotator disagreements.
- Sample size: pilot-informed precision for primary paired difference; statistical collaborator reviews before freeze.

32. Planned figures/tables

- Figure 1: architecture and failure pathway through the checker.
- Figure 2: semantic artifact graph and constrained transform example.
- Figure 3: benchmark taxonomy and sampling strategy.
- Figure 4: unsafe acceptance versus valid acceptance, with uncertainty.
- Figure 5: per-error-category performance and ablations.
- Figure 6: representative connectome failure and output consequences.
- Table 1: prior systems and specific checked properties.
- Table 2: benchmark composition and expert agreement.
- Table 3: computational overhead and held-out generalization.

VI. GOVERNANCE, REPRODUCIBILITY, AND RISKS

33. Ethical and operational safeguards

Use public and synthetic MRI data for primary experiments; do not place PHI or internal institutional data into public agent services. Verify dataset licenses, IRB applicability, university ownership, grant policies, computing-system permissions and export/de-identification rules before involving URMC artifacts. Record and disclose any code and research overlap with Paper 1/SciCap and NeuroInsight. Tool deployment is research use only; it is not validated medical device software.

34. Reproducibility package

- Tagged SciGuard source release, license, CITATION.cff, changelog and versioned schema.
- Reproducible public benchmark with frozen train/dev/test splits, task manifests, mutation seeds and annotation handbook.
- Baseline implementations and native validation versions, pinned containers, hardware and environment descriptions.
- Configuration files for agent evaluation, exact prompts and model identifiers where publishing is permitted.
- Raw result records, result-analysis scripts, CI tests, regression fixtures and figures generated from data.
- Data-use and provenance record including dataset accession/revision/checksum and required citation.
- Public issue tracker, model cards/rule coverage documentation, limitations and known unsupported constraints.

35. Critical methodological risks and mitigations

Risk	Consequence	Mitigation
Gold labels depend on local conventions	Apparent errors may be valid alternatives	Record assumptions; allow multiple valid paths; independent reviewers
Synthetic errors too obvious	Inflated detection metrics	Expert-generated realistic cases and authentic failures where permitted
Hidden metadata unavailable	Unfair rejection or false confidence	Explicit UNKNOWN and policy-driven review
Baseline underconfigured	Unconvincing improvement	Use native validation and document tool versions/configuration
LLM contamination	Misleading generalization	Frozen unseen test sets and external task contributions
Rule overfitting	No cross-workflow transfer	Held-out tools, atlas combinations and error families
Runtime scientific invalidity beyond static checks	Excessive validity claim	Empirical output QC; restrict scope
Safety/privacy errors in public demo	Institutional and clinical risk	Synthetic examples only; local-first, read-only design
Novelty overlap with Paper 1	Salami slicing	Distinct algorithmic claims and different primary endpoints

36. Team and responsibilities

Role	Key responsibility
Research engineer / lead author	SciGuard architecture, implementation, automated tests, experiments, analysis and writing
Neuroimaging methods expert	Scientific contracts, valid workflow reference, atlas/registration interpretation, expert adjudication
Formal methods / software verification collaborator	Constraint model, expressivity, proof boundary, comparison to static typing and constraint solvers

Statistical collaborator	Pre-registration, sample size, clustering, uncertainty and multiple comparisons
External laboratory reviewer	Independent usability, task evaluation and reproducibility audit

37. Approximate cost drivers

Public datasets, standards and most libraries are free to access, subject to individual licenses. Major time/cost drivers are expert labeling, realistic workflow execution, storage, CPU/HPC allocation, cloud or API model calls, independent evaluation and package maintenance. Use offline fixture checks for most benchmark tasks and reserve expensive processing for a small clinically relevant materiality subset. No dollar estimate is offered without a real resource budget.

VII. MANUSCRIPT DESIGN AND JOURNAL STRATEGY

38. Recommended manuscript narrative

18. Introduction: distinguish command validity from scientific compatibility; provide a concrete failure example and contribution statement.
19. Related work: Nipype, Boutiques, BIDS, CWL, engines, formal checking and agentic scientific workflows; avoid strawman baselines.
20. Method: artifact semantics, partial evidence, constraint operators, checker, policies, and implementation.
21. Benchmark: task sources, splits, mutations, adjudication and expected outcomes.
22. Results I: known-invalid rejection, accepted-valid coverage and unknown/abstention behavior.
23. Results II: effect on agent-composed plans and repair reliability.
24. Results III: generalization, overhead, and selected downstream connectomics materiality.
25. Discussion: practical deployment and precise limits of claims, metadata integrity, error taxonomy and generalization.
26. Methods supplement: reference rule catalog, provenance, statistical plan, complete ablations, task details and reproducibility.

39. Abstract scaffold (must replace placeholders with measured outcomes)

Scientific workflow engines ensure executable compositions but may not establish the validity of domain assumptions governing artifact compatibility. We introduce SciGuard, a domain-semantic verification framework that represents neuroimaging artifacts, transforms, and pipeline operations through evidence-linked scientific constraints. We construct an expert-reviewed corpus of [N] valid, invalid and indeterminate workflow configurations, and compare SciGuard with conventional validation, tool-native checks, and language-model review. Across [held-out settings], the method [primary measured effect with CI] while preserving [valid acceptance performance]. We demonstrate the implications for [concrete structural connectomics analysis] and release a reproducible tool and evaluation corpus. Results delimit where semantic contracts can improve scientific workflow assurance and where expert/image-level verification remains necessary.

40. Journal tiers: conditional reach

Tier / target	Editorial fit	What must be demonstrated
Stretch: Nature Methods [15,16]	Generalizable life-science method with strong practical impact	Major methodological innovation; strong comparisons and reproducibility; significant biological application; multiple settings/biomedical workflows; robust expert assessment

Stretch: Nature Computational Science [17]	New general computational framework	Formal and cross-disciplinary method, clear computational novelty, broad transfer, strong open benchmark and performance evidence
High: Nature Communications [18]	Important specialist advance with broader interest	Large robust study, material scientific consequences, independent replication and clear methodological significance
High focused: IEEE TMI [19]	Novel medical imaging analysis method	Imaging-specific algorithmic innovation and strong validated quantitative MRI downstream relevance, not only workflow middleware
Strong focused: Imaging Neuroscience [20]	Major neuroimaging method/software tool	Independent real-data use cases; comprehensive validation; open code and sample data; strong relevance to imaging research
Software fallback: SoftwareX / JOSS	Reusable, documented scientific software	Useful implementation, tests, docs and independent adoption; cannot replace methods paper results

41. Nature Methods submission-specific notes

Official scope emphasizes novel methods, strong validation, comparisons with available approaches, general applicability and an important biological application. Current Article guidance indicates an abstract up to 150 words, typically 3,000 main-text words (up to 5,000 with editorial discretion), up to six main display items, and detailed online Methods. These rules can change; verify immediately before submission. Required materials include manuscript, cover letter, references, data and code statements, ethics as applicable, and disclosure of AI use. [15,16]

42. What would justify the highest tier?

- A rigorous semantic representation whose contribution survives comparison with existing typed workflow engines and domain ontologies.
- High scientific-materiality error reduction, not just more warnings; quantified avoided invalid inference or connectome distortion.
- Cross-site datasets, independent investigators and at least a second biomedical domain if claiming broad life-science methods relevance.
- Realistic external tasks, independently curated gold standards and held-out tools.
- Public artifacts allowing reviewers to rerun all key comparisons and reproduce illustrative biological results.
- Consistent gains over strong and fairly configured native validators and LLM baselines, with tractable false positives.

43. If results are narrower, submit appropriately

If SciGuard works only within a tightly curated DWI connectome workflow and mainly delivers engineering quality improvements, Imaging Neuroscience or a focused software/methods venue is more credible than Nature Methods. If results reveal significant generalizable constraint-checking principles across scientific workflow domains, Nature Computational Science becomes plausible. Journal selection should be revisited after pilots and a completed novelty matrix—not fixed in advance.

VIII. ROADMAP, DELIVERABLES AND REFERENCES

44. Stage-gated research plan

Stage	Work	Exit condition

0. Literature + audit (weeks 1–3)	Systematic novelty matrix, access/IP check, exact baseline list	Written gap statement and preregistration draft
1. MVP (weeks 4–7)	Rule schema, CLI, 20–40 fixtures, deterministic reports	Tests pass; external expert checks sample outputs
2. Pilot (weeks 8–11)	40–60 tasks, reviewers, first baselines, analysis	Estimate error rates, disagreement and task coverage
3. Frozen study design (weeks 12–14)	Power/precision, primary endpoint, splits, holdouts	Signed protocol and frozen rule/test boundary
4. Main study (months 4–7)	Expanded dataset, paired baselines, ablations, uncertainty	All prespecified runs complete; robust audit
5. Real-data materiality (months 6–8)	Execution of curated subset, imaging QC, downstream effects	Validated case studies and transparent limitations
6. Manuscript/release (months 8–10+)	Results, peer internal review, v1 release and submission	Reproducible figures and journal-specific package

Timing is an illustrative project plan, not a guaranteed completion estimate. External review, approvals, compute and negative results may change sequencing.

45. Go/no-go criteria

- GO: Pilot detects an important error class not already handled adequately by configured native validators.
- GO: Expert adjudicators can agree on validity/unknown labels with a defensible protocol.
- GO: False rejection rate is low enough for useful deployment, or calibrated abstention preserves safe operation.
- GO: Results persist on at least some held-out tool/workflow combinations.
- REVISE: Gains vanish under strong baselines, annotations are inconsistent, or almost all cases are UNKNOWN.
- REPOSITION: If methodological novelty is weak but tool is useful, publish a narrower software or applied validation study.

46. First 20 actionable engineering/research tasks

27. Read Nipype and Boutiques technical descriptions closely and map their validation capabilities
28. Catalog BIDS/derivatives and transform conventions relevant to connectomics
29. Document one complete DKT-connectome data lineage without restricted data
30. Define artifact type vocabulary and canonical units/spaces
31. Define status semantics PASS/FAIL/UNKNOWN/NEEDS_REVIEW
32. Draft 15 domain constraints with citations and known limits
33. Create source-linked tool contracts for five operations
34. Build Pydantic models and schema validation
35. Implement graph edge checks
36. Implement provenance and transform-direction predicate
37. Implement atlas ID and label-table predicate
38. Implement discrete-label interpolation predicate
39. Implement command-line report and JSON schema
40. Generate 20 known-valid and 20 targeted-invalid fixtures
41. Create 10 incomplete-evidence UNKNOWN fixtures
42. Recruit two independent expert annotators and establish adjudication rubric
43. Choose and pin baseline versions
44. Pilot LLM review with equal metadata access
45. Draft preregistration and statistical-analysis plan
46. Open permitted code/documentation and create DOI-ready release checklist

47. Readiness inventory (honest assessment)

Asset	Present basis	Needs verification or creation
Neuroimaging engineering knowledge	Substantial prior pipelines and connectomics experience	Independent domain annotation
DKT/structural-connectome chain	Existing engineering experience	Permission to reuse code; audited technical contracts
NeuroInsight platform	Prior design and implementation	Modular integration point and test maturity
HPC infrastructure	Operational experience	Approved research allocations, reproducible containers
Data	Public sources identified	Exact snapshots, processing feasibility and licenses
Semantic checker	Proposed; not confirmed built	Full implementation, correctness tests, profiling
Benchmark	Proposed; not confirmed built	Gold labels, task splits, public release
Peer-reviewed results	Not available	Controlled experiments and evaluation

48. Selected verified bibliography and standards

[1] Gorgolewski KJ et al. (2011). Nipype: a flexible, lightweight and extensible neuroimaging data processing framework in Python. Front Neuroinform. DOI:10.3389/fninf.2011.00013.
https://pubmed.ncbi.nlm.nih.gov/21897815/

[2] Gorgolewski KJ et al. (2016). The Brain Imaging Data Structure, a format for organizing and describing outputs of neuroimaging experiments. Sci Data. DOI:10.1038/sdata.2016.44.
https://www.nature.com/articles/sdata201644

[3] Gorgolewski KJ et al. (2017). BIDS Apps: Improving ease of use, accessibility, and reproducibility of neuroimaging data analysis methods. PLoS Comput Biol. DOI:10.1371/journal.pcbi.1005209.
https://doi.org/10.1371/journal.pcbi.1005209

[4] Glatard T et al. (2018). Boutiques: a flexible framework to integrate command-line applications in computing platforms. GigaScience. DOI:10.1093/gigascience/giy016.
https://pubmed.ncbi.nlm.nih.gov/29718199/

[5] Common Workflow Language. User Guide and specification. Living standard; version to be pinned at experiment freeze.
https://www.commonwl.org/user_guide/

[6] Di Tommaso P et al. (2017). Nextflow enables reproducible computational workflows. Nat Biotechnol. DOI:10.1038/nbt.3820.
https://doi.org/10.1038/nbt.3820

[7] Koster J, Rahmann S. (2012). Snakemake - a scalable bioinformatics workflow engine. Bioinformatics. DOI:10.1093/bioinformatics/bts480.
https://doi.org/10.1093/bioinformatics/bts480

[8] Model Context Protocol. Tool and resource interface documentation; living specification.
https://modelcontextprotocol.io/specification/

[9] OpenAPI Initiative. OpenAPI Specification; living standard.
https://spec.openapis.org/oas/latest.html

[10] RO-Crate. Specification and research object packaging guidance.
https://www.researchobject.org/ro-crate/specification.html

[11] W3C PROV-O. The PROV Ontology. W3C Recommendation.
https://www.w3.org/TR/prov-o/

[12] de Moura L, Bjorner N. (2008). Z3: An Efficient SMT Solver. TACAS. DOI:10.1007/978-3-540-78800-3_24.
https://doi.org/10.1007/978-3-540-78800-3_24

[13] Miao et al. (2026). Reimagining research papers as interactive and reliable AI agents (Paper2Agent). Nature. Verify bibliographic metadata at final citation lock.
https://www.nature.com/articles/s41586-026-11044-y

[14] Balis et al. (2026). From Research Question to Scientific Workflow: Leveraging Agentic AI for Science Automation. Preprint; peer-review status must be checked.
https://arxiv.org/abs/2604.21910

[15] Nature Methods. Aims and Scope (official journal guidance).
https://www.nature.com/nmeth/submission-guidelines/about/aims

[16] Nature Methods. Content Types and preparing submission (official guidance).
https://www.nature.com/nmeth/content

[17] Nature Computational Science. Aims and Scope (official guidance).
https://www.nature.com/natcomputsci/aims

[18] Nature Communications. Aims and Scope (official guidance).
https://www.nature.com/ncomms/aims

[19] IEEE Transactions on Medical Imaging. Scope and editorial criteria.
https://ieeetmi.org/scope/

[20] Imaging Neuroscience. Guide for authors including Software Toolbox criteria.
https://direct.mit.edu/imag/pages/guide_for_authors

[21] Workflow Run RO-Crate working group. Workflow Run Crate specification v0.5, Zenodo 2024.
https://zenodo.org/records/12159311

[22] Pernet CR et al. OpenNeuro/OpenfMRI ds000114: test-retest fMRI dataset with T1w and diffusion; verify exact data snapshot.
https://openfmri.org/dataset/ds000114

[23] BIDS Standard. Official specification, BIDS derivatives and metadata conventions.
https://bids-specification.readthedocs.io/

[24] BIDS examples GitHub repository. Example data structures; inspect actual voxel data separately.
https://github.com/bids-standard/bids-examples

49. Reference and literature audit tasks before submission

The bibliography is a verified starter collection, not a systematic exhaustive review. Before manuscript submission: (1) validate every DOI, year, author list and current standard revision; (2) search for semantic scientific workflow typing/ontology papers beyond this initial set; (3) include recent 2026 agent workflow verification papers; (4) verify all journal requirements anew; (5) maintain a novelty-claim-to-reference traceability table; (6) follow required licenses and citations for every dataset/tool.

End of research planning document. No experiment or journal acceptance is implied.